THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Some Applications of Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $2+2\sqrt{3}$ m	$h(\cot 30^\circ - \cot 45^\circ) = 4$ gives $h = 4 \div (\sqrt{3}-1)$; Height $= 2+2\sqrt{3}$ m (distances $6+2\sqrt{3}$ m and $2+2\sqrt{3}$ m)
2	**(d)** 20 m	Height of the top of the flagstaff $= d\tan 60^\circ = 30$ m; Length $= 30 - 10 = 20$ m
3	**(a)** 2 minutes	Distance covered in 4 min $= h\sqrt{3} - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}}$; distance left $= \frac{h}{\sqrt{3}}$, which is half of it; Time left $= 4 \div 2 = 2$ minutes
4	**(a)** 12 m	Both objects are on the same side, so the distances are subtracted; Distance $= 12$ m
5	**(b)** 4 m	Part of the tower above the top of the building $= d\tan 30^\circ = 1$ m; Tower $= 4$ m
6	**(a)** $9+3\sqrt{3}$ m	The points are on opposite sides, so the distances add; Distance $= 9+3\sqrt{3}$ m
7	**(c)** 30°	$\tan\theta = \frac{\sqrt{3}}{3}$ which is $\tan 30^\circ$; $\theta = 30^\circ$
8	**(c)** $3+3\sqrt{3}$ m	$h(\cot 30^\circ - \cot 45^\circ) = 6$ gives $h = 6 \div (\sqrt{3}-1)$; Height $= 3+3\sqrt{3}$ m (distances $9+3\sqrt{3}$ m and $3+3\sqrt{3}$ m)
9	**(a)** Both A and R are true and R is the correct explanation of A.	With h fixed, smaller d gives larger $\tan\theta$, hence larger θ; R is the explanation.
10	**(c)** A is true but R is false.	$\tan 60^\circ = \frac{10\sqrt{3}}{10} = \sqrt{3}$ so A is true; R is false because $\tan 60^\circ = \sqrt{3}$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Let height $= h$; the distances from the foot are $h\cot 45^\circ$ and $h\cot 60^\circ$, which differ by 6 m; $h(\cot 45^\circ - \cot 60^\circ) = 6$ gives $h = 6 \div (1 - \frac{\sqrt{3}}{3})$ (1); Height $= 9+3\sqrt{3}$ m (distances $9+3\sqrt{3}$ m and $3+3\sqrt{3}$ m) (1).	1+1
12	Distance of the point from the building: $d = 9\cot 45^\circ = 9$ m; Height of the top of the flagstaff $= d\tan 30^\circ = 3\sqrt{3}$ m (1); Length $= 3\sqrt{3} - 9 = 3\sqrt{3} - 9$ m (1).	1+1
13	Let height $= h$: distances $h\cot 30^\circ = h\sqrt{3}$ and $h\cot 60^\circ = \frac{h}{\sqrt{3}}$; Distance covered in 6 min $= h\sqrt{3} - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}}$; distance left $= \frac{h}{\sqrt{3}}$, which is half of it (1); Time left $= 6 \div 2 = 3$ minutes (1).	1+1
14	Let height $= h$; the distances from the foot are $h\cot 45^\circ$ and $h\cot 60^\circ$, which differ by 12 m; $h(\cot 45^\circ - \cot 60^\circ) = 12$ gives $h = 12 \div (1 - \frac{\sqrt{3}}{3})$ (1); Height $= 18+6\sqrt{3}$ m (distances $18+6\sqrt{3}$ m and $6+6\sqrt{3}$ m) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 18\sqrt{3} = d\tan 30^\circ$ (1); Subtract: $d(\tan 60^\circ - \tan 30^\circ) = 18\sqrt{3}$, so $d = 27$ (1); $H = d\tan 60^\circ = 27\sqrt{3}$; tower $27\sqrt{3}$ m, distance 27 m (1).	1+1+1
16	Distance of the point from the building: $d = 14\cot 45^\circ = 14$ m (1); Height of the top of the flagstaff $= d\tan 60^\circ = 14\sqrt{3}$ m (1); Length $= 14\sqrt{3} - 14 = 14\sqrt{3} - 14$ m (1).	1+1+1
17	Distances of the points from the foot: $h\cot 45^\circ = 6$ m and $h\cot 60^\circ = 2\sqrt{3}$ m (1); The points are on opposite sides, so the distances add (1); Distance $= 6+2\sqrt{3}$ m (1).	1+1+1
	OR: Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 20\sqrt{3} = d\tan 30^\circ$ (1); Subtract: $d(\tan 60^\circ - \tan 30^\circ) = 20\sqrt{3}$, so $d = 30$ (1); $H = d\tan 60^\circ = 30\sqrt{3}$; tower $30\sqrt{3}$ m, distance 30 m (1).	1+1+1
18	Distance of the point from the building: $d = 16\cot 45^\circ = 16$ m (1); Height of the top of the flagstaff $= d\tan 60^\circ = 16\sqrt{3}$ m (1); Length $= 16\sqrt{3} - 16 = 16\sqrt{3} - 16$ m (1).	1+1+1
	OR: Distances of the points from the foot: $h\cot 45^\circ = 12$ m and $h\cot 60^\circ = 4\sqrt{3}$ m (1); The points are on opposite sides, so the distances add (1); Distance $= 12+4\sqrt{3}$ m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

19	(i) $\tan\theta$ (1). (ii) $\sqrt{3}$ (1). (iii) $h = 20\tan 60^\circ = 20\sqrt{3} = 34.6$ m (1). (iv) New distance $= \dfrac{20\sqrt{3}}{\tan 30^\circ} = 20\sqrt{3} \times \sqrt{3} = 60$ m (1); distance moved $= 60 - 20 = 40$ m (1).	1+1+1+2
20	Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 6 = d\tan 45^\circ$ (2); Subtract: $d(\tan 60^\circ - \tan 45^\circ) = 6$, so $d = 3 + 3\sqrt{3}$ (2); $H = d\tan 60^\circ = 9 + 3\sqrt{3}$; tower $9 + 3\sqrt{3}$ m, distance $3 + 3\sqrt{3}$ m (1).	2+2+1
	OR: Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 72 (2); $h\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = 72$ gives $h = 18\sqrt{3}$ (2); Height $18\sqrt{3}$ m; distances 18 m and 54 m (1).	2+2+1

Mistakes to avoid:

- Measuring the angle from the vertical or from the line of sight instead of from the horizontal.
- Using $\tan 30^\circ = \sqrt{3}$ or $\tan 60^\circ = \dfrac{1}{\sqrt{3}}$; the values are the other way round.
- Confusing the angle of depression with the angle at the foot of the tower; use alternate angles and the horizontal at the top.
- Forgetting to add the observer's height (or subtract it) when the angle is measured from the eye level.
- Subtracting the distances when the two observation points are on opposite sides of the tower (they must be added).
- Giving the final answer without units or leaving $\sqrt{3}$ unsubstituted when the question asks for a decimal value.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Some Applications of Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 18 m	Both objects are on the same side, so the distances are subtracted; Distance $= 18$ m
2	**(c)** 12 m	Part of the tower above the top of the building $= d\tan 30^\circ = 3$ m; Tower $= 12$ m
3	**(c)** $15 + 5\sqrt{3}$ m	The points are on opposite sides, so the distances add; Distance $= 15 + 5\sqrt{3}$ m
4	**(b)** 30°	$\tan\theta = \frac{\sqrt{3}}{3}$ which is $\tan 30^\circ$; $\theta = 30^\circ$
5	**(c)** $5 + 5\sqrt{3}$ m	$h(\cot 30^\circ - \cot 45^\circ) = 10$ gives $h = 10 \div (\sqrt{3} - 1)$; Height $= 5 + 5\sqrt{3}$ m (distances $15 + 5\sqrt{3}$ m and $5 + 5\sqrt{3}$ m)
6	**(d)** 36 m	Height of the top of the flagstaff $= d\tan 60^\circ = 54$ m; Length $= 54 - 18 = 36$ m
7	**(b)** 6 minutes	Distance covered in 12 min $= h\sqrt{3} - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}}$; distance left $= \frac{h}{\sqrt{3}}$, which is half of it; Time left $= 12 \div 2 = 6$ minutes
8	**(c)** 24 m	Both objects are on the same side, so the distances are subtracted; Distance $= 24$ m
9	**(d)** A is false but R is true.	The angle of elevation is measured from the horizontal and lies between 0° and 90°, so A is false; R is true (alternate angles).
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$\tan 45^\circ = 1 = \frac{20}{20}$ so A is true; R is also true but it is not the reason for the equal lengths.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Distance of the point from the building: $d = 15\cot 45^\circ = 15$ m; Height of the top of the flagstaff $= d\tan 30^\circ = 5\sqrt{3}$ m (1); Length $= 5\sqrt{3} - 15 = 5\sqrt{3} - 15$ m (1).	1+1
12	Let height $= h$: distances $h\cot 30^\circ = h\sqrt{3}$ and $h\cot 60^\circ = \frac{h}{\sqrt{3}}$; Distance covered in 10 min $= h\sqrt{3} - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}}$; distance left $= \frac{h}{\sqrt{3}}$, which is half of it (1); Time left $= 10 \div 2 = 5$ minutes (1).	1+1
13	Let height $= h$; the distances from the foot are $h\cot 45^\circ$ and $h\cot 60^\circ$, which differ by 16 m; $h(\cot 45^\circ - \cot 60^\circ) = 16$ gives $h = 16 \div (1 - \frac{\sqrt{3}}{3})$ (1); Height $= 24 + 8\sqrt{3}$ m (distances $24 + 8\sqrt{3}$ m and $8 + 8\sqrt{3}$ m) (1).	1+1
14	Distance of the point from the building: $d = 21\cot 45^\circ = 21$ m; Height of the top of the flagstaff $= d\tan 30^\circ = 7\sqrt{3}$ m (1); Length $= 7\sqrt{3} - 21 = 7\sqrt{3} - 21$ m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 22\sqrt{3} = d\tan 30^\circ$ (1); Subtract: $d(\tan 60^\circ - \tan 30^\circ) = 22\sqrt{3}$, so $d = 33$ (1); $H = d\tan 60^\circ = 33\sqrt{3}$; tower $33\sqrt{3}$ m, distance 33 m (1).	1+1+1
16	Distance of the point from the building: $d = 18\cot 45^\circ = 18$ m (1); Height of the top of the flagstaff $= d\tan 60^\circ = 18\sqrt{3}$ m (1); Length $= 18\sqrt{3} - 18 = 18\sqrt{3} - 18$ m (1).	1+1+1
17	Distances of the points from the foot: $h\cot 45^\circ = 18$ m and $h\cot 60^\circ = 6\sqrt{3}$ m (1); The points are on opposite sides, so the distances add (1); Distance $= 18 + 6\sqrt{3}$ m (1).	1+1+1
	OR: Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 24\sqrt{3} = d\tan 30^\circ$ (1); Subtract: $d(\tan 60^\circ - \tan 30^\circ) = 24\sqrt{3}$, so $d = 36$ (1); $H = d\tan 60^\circ = 36\sqrt{3}$; tower $36\sqrt{3}$ m, distance 36 m (1).	1+1+1
18	Distance of the point from the building: $d = 22\cot 45^\circ = 22$ m (1); Height of the top of the flagstaff $= d\tan 60^\circ = 22\sqrt{3}$ m (1); Length $= 22\sqrt{3} - 22 = 22\sqrt{3} - 22$ m (1).	1+1+1
	OR: Distances of the points from the foot: $h\cot 45^\circ = 24$ m and $h\cot 60^\circ = 8\sqrt{3}$ m (1); The points are on opposite sides, so the distances add (1); Distance $= 24 + 8\sqrt{3}$ m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

19	(i) $\sin\theta$ (1). (ii) $60\sin 30^\circ = 30$ m (1). (iii) $60\cos 30^\circ = 30\sqrt{3} = 51.9$ m (1). (iv) New height $= 60\sin 60^\circ = 30\sqrt{3} = 51.9$ m (1); increase $= 51.9 - 30 = 21.9$ m (1).	1+1+1+2
20	Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 10 = d\tan 45^\circ$ (2); Subtract: $d(\tan 60^\circ - \tan 45^\circ) = 10$, so $d = 5 + 5\sqrt{3}$ (2); $H = d\tan 60^\circ = 15 + 5\sqrt{3}$; tower $15 + 5\sqrt{3}$ m, distance $5 + 5\sqrt{3}$ m (1).	2+2+1
	OR: Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 144 (2); $h\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = 144$ gives $h = 36\sqrt{3}$ (2); Height $36\sqrt{3}$ m; distances 36 m and 108 m (1).	2+2+1

Mistakes to avoid:

- Measuring the angle from the vertical or from the line of sight instead of from the horizontal.
- Using $\tan 30^\circ = \sqrt{3}$ or $\tan 60^\circ = \dfrac{1}{\sqrt{3}}$; the values are the other way round.
- Confusing the angle of depression with the angle at the foot of the tower; use alternate angles and the horizontal at the top.
- Forgetting to add the observer's height (or subtract it) when the angle is measured from the eye level.
- Subtracting the distances when the two observation points are on opposite sides of the tower (they must be added).
- Giving the final answer without units or leaving $\sqrt{3}$ unsubstituted when the question asks for a decimal value.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Some Applications of Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 30°	$\tan\theta = \frac{\sqrt{3}}{3}$ which is $\tan 30^\circ$; $\theta = 30^\circ$
2	**(a)** $7 + 7\sqrt{3}$ m	$h(\cot 30^\circ - \cot 45^\circ) = 14$ gives $h = 14 \div (\sqrt{3} - 1)$; Height $= 7 + 7\sqrt{3}$ m (distances $21 + 7\sqrt{3}$ m and $7 + 7\sqrt{3}$ m)
3	**(d)** 48 m	Height of the top of the flagstaff $= d\tan 60^\circ = 72$ m; Length $= 72 - 24 = 48$ m
4	**(a)** 8 minutes	Distance covered in 16 min $= h\sqrt{3} - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}}$; distance left $= \frac{h}{\sqrt{3}}$, which is half of it; Time left $= 16 \div 2 = 8$ minutes
5	**(d)** 30 m	Both objects are on the same side, so the distances are subtracted; Distance $= 30$ m
6	**(c)** 28 m	Part of the tower above the top of the building $= d\tan 30^\circ = 7$ m; Tower $= 28$ m
7	**(c)** $27 + 9\sqrt{3}$ m	The points are on opposite sides, so the distances add; Distance $= 27 + 9\sqrt{3}$ m
8	**(b)** 30°	$\tan\theta = \frac{\sqrt{3}}{3}$ which is $\tan 30^\circ$; $\theta = 30^\circ$
9	**(a)** Both A and R are true and R is the correct explanation of A.	Height $= 10\sin 60^\circ = 10 \times \frac{\sqrt{3}}{2} = 5\sqrt{3}$ m, using R.
10	**(c)** A is true but R is false.	Height $= 100\sin 30^\circ = 50$ m: A true; $\sin 30^\circ = \frac{1}{2}$, so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Let height $= h$: distances $h\cot 30^\circ = h\sqrt{3}$ and $h\cot 60^\circ = \frac{h}{\sqrt{3}}$; Distance covered in 14 min $= h\sqrt{3} - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}}$; distance left $= \frac{h}{\sqrt{3}}$, which is half of it (1); Time left $= 14 \div 2 = 7$ minutes (1).	1+1
12	Let height $= h$; the distances from the foot are $h\cot 45^\circ$ and $h\cot 60^\circ$, which differ by 22 m; $h(\cot 45^\circ - \cot 60^\circ) = 22$ gives $h = 22 \div (1 - \frac{\sqrt{3}}{3})$ (1); Height $= 33 + 11\sqrt{3}$ m (distances $33 + 11\sqrt{3}$ m and $11 + 11\sqrt{3}$ m) (1).	1+1
13	Distance of the point from the building: $d = 27\cot 45^\circ = 27$ m; Height of the top of the flagstaff $= d\tan 30^\circ = 9\sqrt{3}$ m (1); Length $= 9\sqrt{3} - 27 = 9\sqrt{3} - 27$ m (1).	1+1
14	Let height $= h$: distances $h\cot 30^\circ = h\sqrt{3}$ and $h\cot 60^\circ = \frac{h}{\sqrt{3}}$; Distance covered in 22 min $= h\sqrt{3} - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}}$; distance left $= \frac{h}{\sqrt{3}}$, which is half of it (1); Time left $= 22 \div 2 = 11$ minutes (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 26\sqrt{3} = d\tan 30^\circ$ (1); Subtract: $d(\tan 60^\circ - \tan 30^\circ) = 26\sqrt{3}$, so $d = 39$ (1); $H = d\tan 60^\circ = 39\sqrt{3}$; tower $39\sqrt{3}$ m, distance 39 m (1).	1+1+1
16	Distance of the point from the building: $d = 24\cot 45^\circ = 24$ m (1); Height of the top of the flagstaff $= d\tan 60^\circ = 24\sqrt{3}$ m (1); Length $= 24\sqrt{3} - 24 = 24\sqrt{3} - 24$ m (1).	1+1+1
17	Distances of the points from the foot: $h\cot 45^\circ = 36$ m and $h\cot 60^\circ = 12\sqrt{3}$ m (1); The points are on opposite sides, so the distances add (1); Distance $= 36 + 12\sqrt{3}$ m (1).	1+1+1
	OR: Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 28\sqrt{3} = d\tan 30^\circ$ (1); Subtract: $d(\tan 60^\circ - \tan 30^\circ) = 28\sqrt{3}$, so $d = 42$ (1); $H = d\tan 60^\circ = 42\sqrt{3}$; tower $42\sqrt{3}$ m, distance 42 m (1).	1+1+1
18	Distance of the point from the building: $d = 30\cot 45^\circ = 30$ m (1); Height of the top of the flagstaff $= d\tan 60^\circ = 30\sqrt{3}$ m (1); Length $= 30\sqrt{3} - 30 = 30\sqrt{3} - 30$ m (1).	1+1+1
	OR: Distances of the points from the foot: $h\cot 45^\circ = 42$ m and $h\cot 60^\circ = 14\sqrt{3}$ m (1); The points are on opposite sides, so the distances add (1); Distance $= 42 + 14\sqrt{3}$ m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

19	(i) 60° (angle of depression = angle of elevation, alternate angles) (1). (ii) $\frac{60}{\tan 60^\circ} = 20\sqrt{3} = 34.6$ m (1). (iii) $\frac{60}{\tan 30^\circ} = 60\sqrt{3} = 103.8$ m (1). (iv) $60\sqrt{3} - 20\sqrt{3} = 40\sqrt{3} = 69.2$ m (2).	1+1+1+2
20	Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 14 = d\tan 45^\circ$ (2); Subtract: $d(\tan 60^\circ - \tan 45^\circ) = 14$, so $d = 7 + 7\sqrt{3}$ (2); $H = d\tan 60^\circ = 21 + 7\sqrt{3}$; tower $21 + 7\sqrt{3}$ m, distance $7 + 7\sqrt{3}$ m (1).	2+2+1
	OR: Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 192 (2); $h\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = 192$ gives $h = 48\sqrt{3}$ (2); Height $48\sqrt{3}$ m; distances 48 m and 144 m (1).	2+2+1

Mistakes to avoid:

- Measuring the angle from the vertical or from the line of sight instead of from the horizontal.
- Using $\tan 30^\circ = \sqrt{3}$ or $\tan 60^\circ = \frac{1}{\sqrt{3}}$; the values are the other way round.
- Confusing the angle of depression with the angle at the foot of the tower; use alternate angles and the horizontal at the top.
- Forgetting to add the observer's height (or subtract it) when the angle is measured from the eye level.
- Subtracting the distances when the two observation points are on opposite sides of the tower (they must be added).
- Giving the final answer without units or leaving $\sqrt{3}$ unsubstituted when the question asks for a decimal value.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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